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Human Intention Modeling for Physical Human–Robot Co-Manipulation

Control-relevant representations, inference methods,
and an initial experimental study

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Reference task: Collaborative table carrying

Task in Diffusion Co-Policy

- ▶ A human and a robot act at opposite ends of a rigid table.
- ▶ They move the table to a goal while avoiding obstacles.
- ▶ Each partner applies a planar force; success requires coordinated translation and rotation.

Information available online

- ▶ table pose and velocity;
- ▶ initial pose, goal, and obstacle locations;
- ▶ recent human force commands and robot actions;
- ▶ for physical co-manipulation: add measured 6D interaction wrench and human motion.

Why human intention should be modeled

The same table state can lead to different valid routes, rotations, and leader–follower roles. Without predicting the human’s choice, the robot may resist the human, apply conflicting forces, or react too late.

Human intention is hidden but control-relevant

What the robot observes

- ▶ object pose and velocity x_t^O ;
- ▶ interaction wrench $\mathbf{w}_t^{\text{int}}$ (3D force + 3D torque);
- ▶ human/robot motion and recent actions;
- ▶ task geometry, goal, and constraints c .

What remains hidden

- ▶ desired object motion or local goal;
- ▶ leader/follower role and compliance;
- ▶ effort, force, and safety preferences;
- ▶ responses to possible robot actions.

Central question

How should human intention be represented and inferred so that it improves real-time robot planning and control in physically coupled tasks?

Notation used throughout the discussion

Indices and agents

- ▶ t : current time; k : a time inside the prediction horizon;
- ▶ H : number of future steps being predicted;
- ▶ superscripts O , H , and R : object, human, and robot.

Signals

x : state or pose; u : action or force command;
 $\mathbf{w}^{\text{int}}$: measured interaction wrench.

Model inputs

h_t : recent interaction history up to time t ; c : task context, including goal, geometry, object properties, and constraints.

Model notation

$p(a | b)$: distribution of a given b ; ψ : learned model parameters; $*$: desired or latent quantity; $\dot{x}$: time derivative of x . PACE means Peer-Aware Cost Estimation; N-PACE is its nonlinear extension.

A useful model must resolve four uncertainties

- ▶ **Behavioral multimodality**: the same state may admit several valid goals, paths, or roles.
- ▶ **Partial observability**: motion and force provide indirect, sometimes conflicting evidence.
- ▶ **Closed-loop interaction**: the human responds to the robot and may change strategy.
- ▶ **Bounded rationality**: actions may be delayed, noisy, effort-limited, or non-optimal.

Terminology

“Multimodal sensing” means multiple input types (motion, wrench, vision).

“Multimodal behavior” means multiple possible future human strategies.

Shared papers show human models benefit interaction

Diffusion Co-Policy

- ▶ Generates several possible joint force/action sequences
- ▶ Demonstrates adaptation and role switching
- ▶ Provides the physical test setting for collaborative transport

Bounded-Rational Model

- ▶ Shows joint human actions can be represented by a game model
- ▶ Bounded rationality fits observed behavior better
- ▶ Leadership and adaptation affect performance

PACE / N-PACE

- ▶ Estimates intent/cost online while treating the peer as an adapting learner
- ▶ Shows that peer modeling can improve interaction policies
- ▶ Provides a game-theoretic interface for online control

Combined insight

Human intention is modelable and useful for control; the unresolved question is which representation is sufficiently expressive, identifiable, and transferable.

A hierarchy of control-relevant intention representations

High-level layer: interaction primitive

Output: $z_H = (m, g, v, \rho)$, describing the current phase, local goal, speed/force level, and interaction role.



Desired object motion

Terminal pose, path mode, or desired trajectory ξ_H^* .

Cost and preferences

Features ϕ and weights w_H for progress, effort, force, and safety.

Local dynamical system

Human-intended vector field $f_H(x^O; \theta_H)$ over object motion; θ_H denotes its human-specific parameters.

Interactive response distribution

Human wrench/action responses conditioned on candidate robot actions.

How the intention representations will be evaluated

1. Intention identifiability

Can the model distinguish route, role, and task phase early? Measure classification accuracy, posterior confidence, and detection delay.

2. Future prediction

Measure trajectory displacement, wrench prediction error, likelihood of observed behavior, and coverage of different valid futures.

3. Closed-loop control value

Measure task success, completion time, human effort, conflicting interaction force, safety violations, and online computation time.

4. Transfer and adaptation

Test unseen users, objects, obstacle layouts, and tasks; report zero-shot performance and data required for personalization.

High-level option: Reusable interaction primitives

Switching latent-state model

$$z_t^H = (m_t, g_t, v_t, \rho_t), \quad m_t \in \mathcal{M}_{\text{int}}$$

where $\mathcal{M}_{\text{int}}$ includes translate, rotate, hold, align, insert, and yield.

- ▶ m_t : current interaction primitive;
- ▶ g_t : local direction, pose, or contact goal;
- ▶ v_t : desired speed or interaction-force level;
- ▶ ρ_t : leader/follower role or compliance.

Why it supports generalization

Different tasks reuse a small vocabulary: transport may be *translate-rotate-align*; insertion may be *approach-align-hold-insert*.

Initial primitive vocabulary

Start with six object-level modes: translate, rotate, hold, align, insert, and yield. They are defined by distinct motion/force patterns, not by a specific task name.

How primitives are identified and learned

Identify the current primitive

- ▶ Use task-phase labels when the phase is known.
- ▶ Otherwise use Expectation–Maximization (EM) to infer latent switching modes from object motion, wrench, and context.
- ▶ Infer $p(m_t \mid h_t, c)$ online rather than committing to one mode too early.

Decide how many primitives are needed

Begin with the six-mode vocabulary. Merge modes with indistinguishable motion/force responses; add a mode only when repeated residual patterns improve held-out prediction and control.

Learn continuous parameters

Within each mode, train goal, speed/force, and role parameters from future object motion and interaction-wrench prediction.

Option A: Infer the human's objective

Task-conditioned human cost

$$J_H = \sum_{k=t}^{t+H} w_H^\top \phi_\psi(x_k^O, u_k^H, u_k^R, c)$$

Cost and feature notation

J_H : cumulative human cost; ϕ_ψ : learned and physical cost-feature vector; w_H : person-specific human preference weights.

State and action notation

x_k^O : object state; u_k^H : human action; u_k^R : robot action. The sum covers steps $k = t$ through $t + H$ under context c .

Option A: Learning the cost model

Why it is attractive

The weights expose tradeoffs among progress, effort, interaction force, safety, and coordination. The model connects directly to game-theoretic planning, IOC, PACE, and N-PACE.

Training without cost labels

Maximum-entropy inverse optimal control (IOC) learns ϕ_ψ from demonstrated actions; person-specific w_H is inferred as a latent variable.

Main challenges

Different costs can explain the same action; learned features may lose physical meaning; and human behavior is not perfectly optimal.

Practical constraint

Anchor learned features to known physical quantities so that each component of w_H retains a stable interpretation online.

Option A can supply N-PACE's human model

Offline learned representation

- ▶ Train a shared task-conditioned feature map ϕ_ψ from multi-person demonstrations.
- ▶ Learn a context-dependent prior $p(w_H | c)$.
- ▶ Freeze ϕ_ψ at deployment so the online parameter meaning remains stable.

Online N-PACE layer

- ▶ Define $J_H = \sum_k w_H^\top \phi_\psi(x_k^O, u_k^H, u_k^R, c)$.
- ▶ N-PACE estimates w_H and how the human is updating beliefs about the robot.
- ▶ The inferred cost enters the dynamic-game solver to update robot actions.

What the learned model contributes

It replaces task-by-task manual feature design with a reusable cost representation; N-PACE retains peer-aware online inference and interaction-aware control.

Option B: Desired object motion as intention

Desired trajectory model

$$p_{\psi}(\xi_H^* \mid h_t, c), \quad \xi_H^* = \{x_{O,k}^{H,*}\}_{k=t}^{t+H}$$

ξ_H^* is the human-intended object trajectory; $x_{O,k}^{H,*}$ is its desired object state at step k . The model outputs a distribution because several trajectories may be valid.

Possible outputs

- ▶ terminal pose or waypoint;
- ▶ discrete path/strategy mode;
- ▶ distribution over full desired trajectories.

Why it is attractive

Directly useful as a reference for shared-object MPC; easier to label and visualize than latent cost parameters.

Main challenges

Multiple valid trajectories; changing intention; passive prediction ignores robot influence.

Option B: Training and control interface

Training without a unique intent label

Use future object motion as a self-supervised target, but learn several possible future trajectories rather than one “correct” path. Controlled experiments can also provide route or goal labels.

Learning objective

Use a diffusion loss or a loss that keeps the best among several predictions, so left/right routes and alternative rotation strategies are retained instead of averaged together.

Role in the complete system

The desired motion is a low-level reference for shared-object model predictive control (MPC). Combine it with a primitive or preference model when the robot must explain why the human changes strategy.

Related: Townsend et al. (2017); Franceschi et al. (2023).

Option C: What is an attractor model?

Intuitive meaning

An attractor model assumes the human is pulling the shared object toward an internal target. The predicted desired velocity points toward that target and decreases as the object approaches it.

Position example

$$\dot{p}_O^{H,*} = K_p(p_H^* - p_O)$$

p_O : current object position; p_H^* : inferred human target; $K_p > 0$: response gain; $\dot{p}_O^{H,*}$: human-desired object velocity.

Why it is attractive

The robot estimates a target and a few gains instead of assuming that the human solves a complete optimal-control problem.

What the model can and cannot express

It is well suited to local motion toward a goal, but one attractor cannot represent several competing routes or long-term effort/safety tradeoffs.

Option C: Full object-pose model and notation

Human-intended local dynamics

$$\dot{x}_O^{H,*} = f_H(x_O, c; \theta_H), \quad \omega_O^{H,*} = f_{\text{rot}}(R_O, R_H^*; K_{\text{rot}})$$

General model

x_O : current object state; $\dot{x}_O^{H,*}$: human-desired state change; f_H : vector field mapping state and context to desired motion; θ_H : human-specific model parameters.

Rotation model

R_O : current orientation; R_H^* : inferred target orientation; $\omega_O^{H,*}$: desired angular velocity; K_{rot} : rotation gain; f_{rot} : mapping from orientation error to angular velocity.

Difference from Options A and B

Option A models a horizon cost, and Option B predicts a full desired path. Option C predicts the desired velocity at the current state, so it is lighter online but less expressive about long-term tradeoffs.

Option C: Training and online inference

Training without parameter labels

Fit the likelihood of observed object motion using constrained system identification. The desired pose and response gains are latent variables rather than manually labeled quantities.

Switching between local models

Use Expectation–Maximization (EM) offline or a particle filter online to infer θ_H and the active mode from motion and wrench measurements.

Physical constraints

Constrain f_H so the predicted motion remains stable, feasible for the shared object, and consistent with contact and actuation limits.

Related: constraint-aware dynamical-system intent estimation with particle filtering (Shao et al., 2024).

Option D predicts responses to candidate robot actions

Passive prediction

$$p(u_{t:t+H}^H \mid h_t, c)$$

What will the human do next?



Interactive response model

$$p_{\psi}(u_{t:t+H}^H \mid h_t, c, u_{t:t+H}^R)$$

How will the human respond if the robot executes this candidate action?

- ▶ $p(\cdot \mid \cdot)$ is a conditional distribution; h_t is recent history, c is task context, and H is the prediction horizon.
- ▶ u^H : future interaction-wrench/action sequence; u^R : candidate robot action sequence.
- ▶ A diffusion/sequence model can represent multiple responses and role switches.
- ▶ It is predictive, but intention remains implicit unless the model also exposes a goal, trajectory, or preference variable.

Option D versus Diffusion Co-Policy

Diffusion Co-Policy

$$p_{\psi}(u_{t:t+H}^H, u_{t:t+H}^R \mid h_t, c)$$

- ▶ jointly generates human and robot action sequences;
- ▶ directly imitates collaborative demonstrations;
- ▶ captures multimodality and role switching.

Human response model

$$p_{\psi}(u_{t:t+H}^H \mid h_t, c, \tilde{u}_{t:t+H}^R, z_H)$$

- ▶ ψ denotes learned model parameters;
- ▶ evaluates each candidate robot action $\tilde{u}^R$;
- ▶ keeps planning and safety modular;
- ▶ can expose an intent variable z_H .

Tradeoff

The response model supports counterfactual planning, but needs diverse robot-action coverage and calibrated out-of-distribution uncertainty.

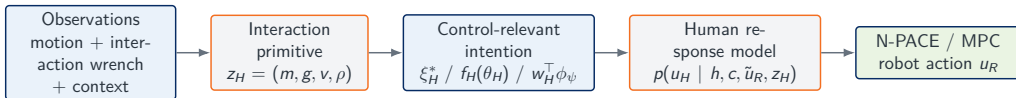
Each representation serves a different modeling layer

Representation	Control interface	Key strength	Main weakness	Best use
Interaction primitive	Latent mode z_H	Cross-task structure	Needs mode discovery	Task-phase inference
Desired trajectory	Reference ξ_H^*	Direct and visual	Misses preferences and robot influence	Shared-object MPC
Cost / preferences	J_H, w_H, ϕ	Interpretable and game-compatible	Identifiability and feature design	Negotiation and tradeoffs
Dynamical system	$f_H(x; \theta_H)$	Lightweight online inference	Limited global multimodality	Local coordination
Response distribution	$p(u_H u_R)$	Flexible and interactive	Intention may remain implicit	Candidate-action evaluation

Proposed position

Use structured intention variables as the control interface, while a learned response model captures multimodality, bounded rationality, and model mismatch.

A simpler hierarchy connects intention modeling to control



Learned offline

Primitive transitions, shared feature/trajectory representations, and the population response prior.

Adapted online

Current primitive, desired motion, preference weights, and person-specific response uncertainty.

Offline priors support online personalization

Offline: learn shared structure

- ▶ task/interaction encoders from multi-person data;
- ▶ trajectory modes and residual cost features;
- ▶ robot-action-conditioned response model;
- ▶ calibrated uncertainty on held-out people.

Online: update compact variables

- ▶ belief over intent mode $b_t(z_H)$;
- ▶ desired motion parameters ξ_H^*, θ_H ;
- ▶ preference weights w_H ;
- ▶ optional small person-specific adapter.

Bayesian personalization from physical interaction

Update equation

$$b_{t+1}(\eta_H) \propto p(y_{t+1} \mid h_t, c, u_t^R, \eta_H) b_t(\eta_H)$$

Belief and latent variables

$\eta_H = (z_H, \theta_H, w_H)$ collects the human primitive, desired-motion parameters, and preference weights. $b_t(\eta_H)$ is the robot's current belief over them.

New physical evidence

$y_{t+1} = (\mathbf{w}_t^{\text{int,obs}}, x_{t+1}^O)$ combines the observed wrench and next object state. u_t^R is the executed robot action; $p(\cdot)$ scores how well each human model explains the evidence.

Interpretation

$\propto$ means: multiply the prior belief by the likelihood of the new observation, then normalize. Models that better predict the human–robot interaction receive more probability.

Data collection supplies weak and self-supervision

Minimum synchronized modalities

- ▶ object pose, velocity, and acceleration;
- ▶ 6D interaction wrench;
- ▶ robot state and commanded action;
- ▶ task goal, geometry, contact configuration;
- ▶ optional human body/arm pose and vision.

Where training signals come from

- ▶ **weak labels**: assigned goal, path condition, task phase, role instruction;
- ▶ **self-supervision**: future object motion and interaction wrench;
- ▶ **preference evidence**: choices under controlled effort/force/path tradeoffs;
- ▶ **latent learning**: EM / IOC likelihood when no direct label exists.

Practical collection plan

Use accessible human–human demonstrations for pretraining, then collect human–robot trials with diverse robot actions for interactive-response identification. A curated multimodal benchmark can itself support a dataset/benchmark paper.

First study: Compare intention representations

Physical test setting

Collaborative object transport with:

- ▶ two feasible paths around an obstacle;
- ▶ translation–rotation tradeoffs;
- ▶ changing leader/follower roles;
- ▶ synchronized object state and 6D interaction wrench.

Model comparison

1. desired-trajectory prediction;
2. fixed-feature IOC / cost inference;
3. switching dynamical-system inference;
4. interactive response model;
5. hierarchical hybrid model.

Central hypothesis

Combining structured intention variables with robot-action-conditioned response prediction will improve early intent inference and closed-loop coordination.

Separate model quality from team performance

Human-model quality

- ▶ path/role classification and detection delay;
- ▶ desired-trajectory displacement error;
- ▶ force/action likelihood and calibration;
- ▶ adaptation after strategy changes;
- ▶ leave-one-person-out generalization.

Closed-loop interaction quality

- ▶ task success and completion time;
- ▶ human and robot effort;
- ▶ unnecessary interaction force;
- ▶ path efficiency and smoothness;
- ▶ safety and comfort violations.

Critical ablations

Motion vs. motion+wrench; passive vs. robot-action-conditioned; fixed vs. learned features; population prior vs. online personalization.

A focused decision can define the first paper

Proposed contribution

Compare control-relevant human intention representations for physical collaborative transport, then combine their complementary strengths in a hierarchical model.

Questions for discussion

- ▶ Which output is most useful for the intended controller: desired object motion, cost/preferences, or future interaction wrench?
- ▶ Which intentions can be manipulated or labeled reliably in the planned testbed?
- ▶ Should the initial paper emphasize model comparison or one hybrid implementation?
- ▶ What robot, object, and force/torque sensing setup is already available?

Recommended starting scope

Begin with desired-trajectory and cost/preference baselines, then add robot-action-conditioned response prediction as the learned component.

Follow-up Research Directions

Longer-term subprojects in sensing, control, and cross-task generalization

Additional notation used in the subprojects

Human-model learning

J_H : human cost; w_t^H : time-varying human preference weights; ϕ_ψ : learned cost features; u_{future}^H and u_{future}^R : future human and robot action sequences.

Probability model

p_ψ : learned conditional distribution; ψ : its trained parameters. The conditioning variables are written after the vertical bar $|$.

Robot planning

u_R^* : selected robot action sequence; J_R : robot planning cost; $\mathbb{E}$: average over possible human responses; $\arg \min$: the robot action that produces the smallest expected cost.

Shared inputs

x_t : system state; h_t : recent history; c : task context. Superscripts H and R denote human and robot quantities.

SP1: Build the physical evidence base

Theme and focus

Physically grounded multimodal data

- ▶ Build shared-object transport experiments
- ▶ Synchronize motion, object state, and force/torque
- ▶ Create ambiguous paths, role switching, and strategy changes
- ▶ Compare motion-only, wrench-only, and multimodal inference

Challenges → solutions

- ▶ No direct intent labels → hierarchical task/strategy/action labels
- ▶ Noisy contact data → filtering and contact-state estimation
- ▶ High person variability → probabilistic latent-state models
- ▶ Costly data → human–human, simulation, and HRI data mixture

Expected paper: A multimodal benchmark and systematic study of force/torque for human intent inference.

SP2: Make intent and preference explicit

Bounded-rational human model

$$J_H = \sum_t w_t^{H^\top} \phi_\psi(x_t, u_t^H, u_t^R, c)$$

Reusable features may represent progress, pose, effort, interaction force, safety, smoothness, and coordination.

- ▶ Infer intent, role, preference, and rationality online
- ▶ Represent the human's belief about robot intent
- ▶ Treat the human as another adapting learner

Challenges → solutions

- ▶ Cost identifiability → Bayesian / maximum-entropy IOC
- ▶ Ambiguous intent → informative robot actions
- ▶ Unknown adaptation → latent learning-state estimation
- ▶ Online computation → local game approximations and MPC

Expected paper: Peer-aware intent and preference inference for physical co-manipulation.

SP3: Learn responses to possible robot actions

Interactive response prediction

$$p_{\psi}(u_{\text{future}}^H \mid h_t, c, u_{\text{future}}^R)$$

- ▶ Predict multimodal human response trajectories
- ▶ Condition on motion, wrench, context, and candidate robot actions
- ▶ Predict uncertainty and latent-state transitions
- ▶ Compare Transformer, diffusion, and latent-variable models

Challenges → solutions

- ▶ Limited action coverage → diverse robot probing
- ▶ Correlation vs. response → counterfactual conditioning
- ▶ Out-of-distribution actions → uncertainty estimation and detection
- ▶ Slow inference → distilled or few-step diffusion

Expected paper: Robot-action-conditioned diffusion models for multimodal human response prediction.

SP4: Close the loop through safe control

Human-model-aware planning

$$u_R^* = \arg \min_{u_R} \mathbb{E}_{u_H \sim p_\psi} [J_R]$$

subject to object dynamics, actuation, force, collision, and safety constraints.

- ▶ Optimize progress, team effort, and interaction force
- ▶ Replan as the human model is updated
- ▶ Use a safety filter before execution

Expected paper: Uncertainty-aware planning and safe control with interactive human response models.

Challenges → solutions

- ▶ Prediction error → short-horizon MPC
- ▶ Distributional complexity → scenario sampling
- ▶ Excess conservatism → risk-sensitive objectives
- ▶ Hard safety limits → control-barrier safety filter

SP5: Transfer human concepts across tasks and people

Task-conditioned generalization

Shared human latent concepts:

- ▶ intent and local strategy;
- ▶ leader–follower role and compliance;
- ▶ effort, safety, and interaction preferences;
- ▶ reaction delay and adaptation state.

Encode goals, object dynamics, geometry, contacts, and constraints as task context. Adapt only latent states, preference weights, or a small adapter online.

Expected paper: A task-conditioned human model for cross-task co-manipulation and online personalization.

Challenges → solutions

- ▶ Different task structures → object-centric states and wrenches
- ▶ Limited physical data → simulation and human–human pretraining
- ▶ Negative transfer → structured task encoders
- ▶ Unstable personalization → Bayesian updates or small adapters

Five subprojects answer one connected question chain

SP1 — Observe: What physical signals reveal human intent and interaction strategy?



SP2 — Infer: What does the human intend, prefer, and believe about the robot?



SP3 — Predict: How will the human respond to candidate robot actions?



SP4 — Act: How should the robot coordinate safely under prediction uncertainty?



SP5 — Transfer: What representations generalize across people and tasks?

A focused transport study can test the core hypothesis

Initial task

A human and robot transport a shared object through an environment with multiple feasible paths.

Model inputs

- ▶ Object and human motion
- ▶ Interaction force/torque
- ▶ Goal and environment geometry
- ▶ Candidate robot actions

Comparisons

1. Reactive control without a human model
2. Passive prediction from interaction history
3. Structured intent inference
4. Robot-action-conditioned prediction
5. Hybrid structured + learned model

Evaluation

- ▶ Prediction accuracy and calibration
- ▶ Success, time, and team effort
- ▶ Interaction force and disagreement
- ▶ Adaptation, safety, and comfort violations

The program connects human understanding to robot action

1. A physically grounded multimodal benchmark
2. Bounded-rational, peer-aware intent inference
3. Interactive prediction of responses to robot actions
4. Safe planning over human response distributions
5. Cross-task generalization and online personalization

Long-term vision

A versatile human model that connects physical interaction, human adaptation, and robot decision-making across co-manipulation scenarios.

Proposed starting point

Use a physical collaborative-transport task to test whether force/torque sensing and robot-action-conditioned prediction improve intent inference, coordination, and safe control.

Questions to refine the research direction

- ▶ Which physical co-manipulation task is the most useful initial test case?
- ▶ What sensing and robotic platforms are currently available?
- ▶ Should the first contribution emphasize human intent inference, interactive response prediction, or planning and control?
- ▶ Which components of RISE Lab's existing game-theoretic framework can be extended directly to physical coupling?
- ▶ What level of cross-task generalization would be most valuable for the current project?

Discussion objective

Identify a focused first study that contributes to the larger goal of a generalizable and interactive human model.